

Laboratory Manual: Notebook 2 — Working with Larger Research Datasets

1. Introduction

This laboratory manual is a core component of the "Scientific Workflows for Brain and Behavioral Research" initiative, supported by the National Science Foundation (Award No. OAC-2417875). **The Workflow Designer Role** As the Workflow Designer for this initiative, you are responsible for the architectural viability of the analysis. As research scales from single-participant studies to institutional-level group analyses, your primary challenge shifts from individual calculation to managing data volume and resource allocation. A designer must ensure that the workflow remains stable and does not exceed the hardware limitations—specifically working memory—of the computing environment. **The Methodology: Ask → Build →**

Document To manage the transition to larger datasets, you will utilize the **Ask → Build → Document** cycle. When utilizing AI tools for code generation:

1. **Ask:** Provide precise prompts regarding memory efficiency and scaling.
2. **Build:** Execute the generated code while actively monitoring resource consumption.
3. **Document:** Direct the AI to explain every data-loading step, focusing on the specific amount of memory being allocated. Responsible AI use in a research context requires prioritizing memory efficiency as a pre-flight requirement for system stability.

2. Key Terminology

Term, Definition (Reference Guide Verbatim)

Processor Core, "A single computing unit inside a central processor (CPU). Most modern computers have multiple cores, allowing several tasks to run at the same time."

CPU vs. GPU, "A CPU handles complex, varied tasks one at a time across a few powerful cores. A GPU handles simple, repetitive tasks across thousands of smaller cores simultaneously — useful for image processing and machine learning."

Simultaneous Processing, "Splitting a large task — such as analyzing 50 participants — into smaller chunks that run at the same time across multiple processor cores, rather than one after another."

Automated Data Download/API, "A method of downloading data directly inside a script using a standardized web link, rather than clicking through a website manually. This makes data access reproducible and audit-able."

Working Memory (RAM), RAM (working memory) holds data only while the computer is on — it is extremely fast but temporary.

Optimized Data Files, Saving data as plain text (such as CSV) is easy to read in a spreadsheet but very slow for large datasets. Optimized formats (such as HDF5 or NumPy binary files) are much faster to load and take up less disk space.

3. Setup

To record your progress and ensure your environment is correctly configured for scaling, complete the following profile synchronization. **Set Up Your Profile**

1. Locate the profile synchronization cell in your notebook.
2. Replace the placeholder strings in the code block below with your specific information.

3. Run the cell to record your session.

```
# PROFILE SYNC
student_name = "Your Name"
institution = "Your University/Organization"
active_notebook = "Notebook 02 – Larger Datasets and Simultaneous Processing"

print(f"✅ Profile synced for {student_name} ({institution})")
print(f"Current notebook: {active_notebook}")
```

4. Module 1: Handling Group Datasets (Part A)

Objective This module transitions your workflow from single-participant analysis to group-level datasets. From an architectural perspective, memory management is a pre-flight requirement for stability, while automated data acquisition ensures the reproducibility of the research pipeline. **Activity A.1 — Building a Group Dataset** Use the following prompt to generate the simulation code: "Act as a neuroscience data analyst helping a researcher scale up from a single participant to a full study. Write a Python script using NumPy that simulates loading and combining data from 50 research participants. The script should:

1. For each of 50 participants, generate a data array representing 64 recording channels measured across 1000 time points in 3 experimental conditions.
2. Combine all 50 individual participant arrays into a single group-level dataset efficiently, without creating unnecessary copies of the data.
3. Report the size and shape of the combined dataset and calculate how much working memory it uses — then estimate how large the dataset would become if the study included 500 participants instead. Add plain-language comments explaining what each step does and what the numbers represent. *Note: As the Workflow Designer, it is your responsibility to audit the AI's memory estimations for the 500-participant scale-up. Validate these calculations against **Concept 29 (Memory Overflow)** to ensure the architectural stability of your local or cloud environment.* **Activity A.2 — Downloading Data Reliably** Manual downloads are inherently inferior to automated, resilient systems. Automated systems ensure reproducibility and can recover from network instability without human intervention. Use the following prompt: "Act as a research workflow instructor. Internet connections sometimes drop or get temporarily blocked when downloading large research datasets. Write a Python script that simulates a reliable data download system. The script should:
 4. Simulate a download function that requests 10 chunks of data from a research repository, one at a time.
 5. Randomly cause some of the download attempts to fail, to simulate real network interruptions.
 6. Automatically retry failed downloads, waiting a little longer between each attempt (e.g., wait 1 second after the first failure, 2 seconds after the second, 4 seconds after the third) rather than giving up immediately.

7. Print a clear log showing which downloads succeeded, which failed, and how many retries were needed. Use plain-language comments explaining why this retry approach is better than simply stopping when an error occurs."**Part A Reflection**
 - **What I observed:** Looking at the dataset sizes from Activity A.1, how does memory usage change as the number of participants grows? What practical limits does this create for a research study?
 - **Applying Concepts:** How might **Downsampling (Concept 16)** alleviate the memory growth limits you calculated for 500 participants? Furthermore, how do **Continuous Recordings (Concept 11)** compare to event markers in terms of storage impact?
 - **Connecting to key concepts:** These activities ground the application of **Concepts 11, 14, 16, and 43**. Describe why a reliable retry system for downloads matters for reproducibility in large studies.

5. Module 2: Making Analyses Faster (Part B)

Objective Parallel processing and efficient data formatting are scaling requirements for institutional-level research. These strategies optimize the workflow by distributing computational load and minimizing the I/O bottleneck.**Activity B.1 — Sequential vs. Simultaneous Processing** Use the following prompt to compare processing methods:"Act as a computing performance instructor. Write a Python script that demonstrates the difference between running a data analysis one participant at a time versus running many participants simultaneously using multiple processor cores. The script should:

1. Define a data analysis function that performs a moderately demanding calculation — such as computing a rolling average across a large array.
2. Apply this function to 10 separate datasets one at a time, measuring how long the whole process takes.
3. Apply the same function to the same 10 datasets simultaneously using Python's multiprocessing library, distributing the work across available processor cores, and measuring how long that takes.
4. Display a bar chart comparing the two approaches and calculate the speedup factor. Use plain-language comments explaining what is happening and why one approach is faster."**Activity B.2 — Saving Data in Efficient Formats** Use the following prompt to compare the architectural trade-offs of different storage formats:"Act as a data management instructor. Write a Python script that demonstrates the difference between saving research data as a plain text file versus a more efficient binary format. The script should:
5. Generate a simulated dataset representing measurements from 100,000 time points across multiple recording channels.
6. Save this dataset in two formats: a plain text CSV file (like you would open in a spreadsheet) and an optimized binary file (such as NumPy's .npy format or HDF5).
7. Compare the file sizes on disk for both formats.
8. Measure and compare how long it takes to load each file back into memory for analysis. Use plain-language comments explaining the trade-offs between the two formats and when you would choose each one."**Part B Reflection**

- **What I observed:** From Activity B.1, how much faster was simultaneous processing compared to one-at-a-time? From Activity B.2, how much larger was the plain text file compared to the efficient format?
- **Connecting to key concepts:** Activities B.1 and B.2 demonstrate **Concepts 27, 28, 42, and 43** . Explain in plain language when it makes sense to use simultaneous processing and when the overhead might not be worth it.

6. Integration

A professional research workflow is built upon the dual pillars of **System Stability** and **Research Throughput** . By prioritizing memory management (Part A), you ensure that your architecture is viable and will not suffer a crash as data volume grows. By implementing parallel processing and optimized storage (Part B), you maximize throughput, ensuring the analysis is efficient enough to provide results in a reasonable timeframe. Combining these practices allows for the robust handling of the multi-participant datasets that define modern neuroscience.**Looking Ahead** The next stage of this initiative introduces accessing **National Research Computing Resources (FABRIC)** —creating an account, getting approved on a project, and connecting securely to a shared research computing system to provide the power necessary for even larger-scale workflows.

7. Reference Checklist

Concept Mapping Table

Concept Number,Concept Name,Definition (Reference Guide Verbatim)

11,Continuous Recordings vs. Event Markers,A continuous recording captures an uninterrupted time series of measurements. Event markers are specific timestamps that label when something important happened — such as when a stimulus appeared or a button was pressed.

14,Automated Data Download (API),"A method of downloading data directly inside a script using a standardized web link, rather than clicking through a website manually. This makes data access reproducible and audit-able."

16,Downsampling,"Reducing the number of data points per second in a recording, to save memory and processing time, when the extra detail is not needed for the analysis."

27,Processor Core,"A single computing unit inside a central processor (CPU). Most modern computers have multiple cores, allowing several tasks to run at the same time."

28,CPU vs. GPU,"A CPU handles complex, varied tasks one at a time across a few powerful cores. A GPU handles simple, repetitive tasks across thousands of smaller cores simultaneously — useful for image processing and machine learning."

42,Text Files vs. Optimized Data Files,Saving data as plain text (such as CSV) is easy to read in a spreadsheet but very slow for large datasets. Optimized formats (such as HDF5 or NumPy binary files) are much faster to load and take up less disk space.

43,Simultaneous Processing,"Splitting a large task — such as analyzing 50 participants — into smaller chunks that run at the same time across multiple processor cores, rather than one after another."

Self-Check: What Did You Learn?

Concept Area,Concept Number,Where You Demonstrated It

Neuroscience,Concept 11 — Continuous recordings vs. event markers,Enter Cell ID
Neuroscience,Concept 14 — Automated data download,Enter Cell ID
Neuroscience,Concept 16 — Downsampling,Enter Cell ID
Computing,Concept 27 — Processor cores,Enter Cell ID
Computing,Concept 28 — CPU vs. GPU,Enter Cell ID
Computing,Concept 43 — Simultaneous processing,Enter Cell ID